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Intervention

Telephone peer-delivered intervention for diabetes motivation and support: The telecare exploratory RCT

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ABSTRACT

Objective: To test trial design issues related to measuring the effectiveness of a peer telephone intervention to enhance self-efficacy in type 2 diabetes; evaluate the impact on self-efficacy and clinical outcome; and describe patient and peer experience.

Methods: Eligible patients had raised HbA1c (initial threshold >8%, reduced to >7.4% mid-way through trial). Patients were recruited from 40 general practices and randomised (40:40:20 ratio) to receive routine care alone or, in addition, motivational telephone support from a peer supporter or a diabetes specialist nurse (9 peers and 12 DSNs) for a period of up to 6 months. The primary outcome measure was self-efficacy score, and secondary outcome measures included HbA1c. Patient and telecare supporter satisfaction and experience were evaluated.

Results: In all, 231 patients participated. At 6 months there were no statistically significant differences in self-efficacy scores ($p = 0.68$), HbA1c ($p = 0.87$) or other secondary outcome measures. There was evidence of a high level of acceptability, but peer telecare support was less highly valued than that from a DSN. Some patients stated that they would have valued more information and advice.

Conclusions: Further consideration needs to be given to the targeting of the telecare peer support, its intensity, the training and ongoing supervision of peer supporters, and the extent to which information and advice should be incorporated.

Practice implications: While some patients with poorly controlled type 2 diabetes value peer telephone support, this approach appears not to suit all patients. Further intervention development and evaluation is required before widespread adoption can be recommended.

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1. Introduction

Around 200 million people worldwide have diabetes, and in line with the obesity epidemic this figure is set to increase substantially [1]. Effective self-management may improve diabetes health and prevent or delay the onset of complications [2,3,4], but is difficult and complex to achieve [5,6]. It can be encouraged by enabling patients to understand their condition, combined with resolution of obstacles to adherence [7].

A number of diabetes self-management training interventions exist [7,9,10], and an effective time to intervene may be when a therapeutic change is occurring [8]. Peer support and mentorship may also be important, and in the UK the value of patient experience and knowledge in the provision of effective education and support is being promoted by health policy [11]. There is increasing interest in peer-led and 'expert patient' type interventions [12,13,14]. These have been applied to disease-related, illness-prevention and health promotion topics [15,16,17,18] utilising a range of approaches (e.g. individual one-to-one sessions, self-help/support groups, online computer-mediated groups, telephone) based on the concept of sharing mutual experience [17] and 'experiential knowledge' [19]. Offering support may benefit the peer, as well as the person being helped, resulting in improved physical health [20,21]; increased feelings of self-worth [4,22,23,24]; reduced depressive symptoms [22,25]; improved communication skills [23]; and changes in self-management behaviours [12].

A potential method of promoting effective self-management is through telephone contact. A recent Cochrane review reported the acceptability of telecommunication for patient care [26]. Such contact is already common in diabetes specialist nurse practice in the UK, and may positively affect adherence to self-management

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activities [27], enhance autonomy and confidence in managing diabetes [28], and enable adjustment of diabetes management goals [29,30]. In a recent study conducted in Greater Manchester, individuals living with inadequately controlled type 2 diabetes (HbA1c >7%) were provided with telephone support delivered by trained lay telephonists supported by a diabetes specialist nurse for a period of up to 1 year. Compared with usual care, they had a significant improvement in glycaemic control [31]. The intervention was highly acceptable [32]. However, the study involved only one specialist nurse, so limiting its generalisability.

The telephone intervention tested in the study reported here was designed to allow comparison of peer and nurse support for patients with inadequately controlled type 2 diabetes. Following the recommendations of the UK Medical Research Council (MRC) framework for the development and evaluation of complex interventions to improve health [33], a theoretical basis for the intervention was established drawing on Bandura's self-efficacy theory and the principles of motivational interviewing [34,35]; feasibility work was conducted to develop an understanding of the intervention and its potential outcomes; and finally a pilot study was undertaken [36]. The intervention was intended to increase self-efficacy in relationship to lifestyle behaviours and medication adherence, leading to improvements in clinical outcomes [35].

The current study was an exploratory randomised controlled trial (RCT) undertaken to test methodologies and generate hypotheses rather than produce evidence of a confirmatory nature. Its aims were to test trial design issues related to measuring the effectiveness of a peer telecare support to enhance self-efficacy in type 2 diabetes; to evaluate its impact on self-efficacy and clinical outcome against nurse telecare provision or routine care alone; and to describe patient and peer experience.

2. Methods

Patients were recruited from general practice clinics and allocated to one of three groups: telecare support provided by peer supporters (PS), telecare support provided by diabetes specialist nurses (DSNs), and control group (CG) receiving routine care only. The full protocol has been reported elsewhere [35], and is summarised below. The study received ethical approval from Warwickshire Research Ethics Committee (reference number: REC 610).

2.1. The telecare intervention

Telecare support was intended to supplement routine care by motivating adherence to the advice provided by the general practitioner or practice nurse at the time of a change (medication and/or lifestyle) in the patient's diabetes care. For patients in the intervention arms of the trial, the first telecare call was made 3–5 days later, and the 'standard package' offered subsequent contact at the following points: days 7–10, 14–18, 28–35, 56–70, 120–150. In this way more intense reinforcement of the therapeutic or behaviour change occurred during the early weeks following its initiation, with gradual tapering off of support over a period of months. The frequency of calls, however, was intended to be tailored to patients' individual needs and telecare supporters were taught to negotiate the time of subsequent contact as part of the closure of each call. Telephone record sheets were maintained throughout to ensure consistency of documentation regarding call content, goal setting and achievement, and length of calls.

DSNs were recruited using the DSN directory and peers were recruited through Warwick Diabetes Care User Group and an e-mail support group hosted by Trefoil Solutions. Interested individuals were provided with a role specification. Following an interview in which they were asked about their experience of telephone advice, counselling, and their reasons for wishing to participate in the study, they attended a 2-day training programme developed for the study. The course focused on empowerment, motivational interviewing, and active listening skills, and had an emphasis on role play and practising telephone communication skills [35]. Participants on the course were provided with books on diabetes [37] and on facilitating behaviour change [38], and were encouraged to read particular sections prior to the training to establish a common knowledge base.

Two peer supporters were excluded from the trial following training due to difficulties in mastering the techniques taught. The remaining nine comprised four males and five females, age range 35–75 years (median = 59 years), and diabetes duration range 5–28 years (median = 10 years). Six had type 2 diabetes. The 12 DSNs were all female, age range 35–63 years (median = 42 years) and their diabetes nursing experience ranged from 6 to 22 years (median = 16 years); one had type 1 diabetes.

The telecare supporters were intended to make telephone calls from a confidential space in their workplace or homes. They were paid a small fee reflecting the time commitment involved. They had access via telephone and email to an experienced DSN educationalist with a particular interest in self-management. They were encouraged to contact her about any problems associated with the study. They were also invited to 6-monthly review meetings at which difficult or challenging cases were shared and discussed.

Patients in the control group received a single call from a researcher at day 3–5. The patients were informed that they were allocated to the routine care group and were encouraged to follow the advice of their GP or practice nurse.

2.2. Practices and patients

Recruitment took place between December 2004 and April 2006. In all, 114 general practices within a socio-economically and geographically diverse region in central England were invited to participate, and 43 (37.7%) agreed to do so. Patients with inadequate glycaemic control (raised HbA1c level) were identified during routine diabetes review visits, as well as from practice registers. They were provided with written information about the study and invited to attend an appointment with the practice nurse. The intention was that recruitment would occur systematically within routine practice. Patients who were prescribed a change in treatment or given advice about behaviour change were eligible for inclusion. Baseline data were collected prior to patient randomisation.

2.2.1. Inclusion/exclusion criteria

Eligible patients: Patients with a recent HbA1c greater than 8.0% (to promote patient recruitment this was reduced during the study to HbA1c >7.4% [33]) and been advised at their most recent appointment of the benefit of reducing their glycated haemoglobin through behaviour change (*diet and/or exercise*), with or without a change in prescribed tablet-based therapy.

Exclusions: Patients on insulin, lacking a telephone or adequate English to communicate over the phone, and those with severe accompanying disorders (e.g. mentally ill; severe learning difficulties; severe hearing difficulties) judged by the GP as likely to interfere with outcome interpretation. Patients on insulin were excluded from the trial because not all peer supporters had personal experience (i.e. experiential knowledge) of this treatment, and in the pilot phase of the study it had been found that some general practitioners had concerns about potential risks associated with peer support for such patients [36].

2.3. Randomisation

For patients that consented to participate, a trials coordinator carried out randomisation using opaque sealed envelopes. Owing to patient recruitment taking substantially longer than had been anticipated, which largely reflected the impact of changes in general practitioners' contractual status and system of remuneration which had occurred shortly before the commencement of data collection, the randomisation protocol was adjusted during the study [35]. The revised randomisation schedule (40:40:20 ratio for the patients randomised to the control (CG), peer support (PS) and diabetes specialist nurse (DSN) groups, respectively) was introduced to increase the numbers of patients in the peer telecare support arm of the trial, as this was the main focus of the study.

2.4. Outcome data

The primary outcome was self-efficacy, as measured by the Diabetes Management Self-Efficacy Scale (DMSES) [39]. Secondary outcomes included glycated haemoglobin (HbA1c), cholesterol, *body mass index* (BMI) collected from clinical records and diabetes distress measured by The Problem Areas in Diabetes Scale (PAID) [40]. Data were collected at baseline, 3, and 6 months for the intervention groups, and at baseline and 6 months for the control group. Blood specimens for HbA1c assessment were taken according to local protocols either by practice staff or the local phlebotomy service. They were analysed by the local laboratory services routinely used by the practices, blinded to patient group allocation. No changes in standardisation occurred during the data collection period. For patients in the intervention groups, satisfaction with the telecare intervention was assessed at 6 months using a non-validated questionnaire.

2.5. Sample size calculation

The sample size was based upon the primary outcome, measured using the DMSES and was constructed to detect a difference in the primary outcomes of self-efficacy perceptions and psychological adjustment. A sample of 100 patients per arm gives a power of about 80% to detect a statistically significant difference between two groups in self-efficacy scores with a standardized effect size of 0.35 at a one-sided significance level of 5% [39]. A one-sided level was chosen as the primary interest was in demonstrating that the peer support telecare intervention was superior to the control. Allowing for an attrition rate of 20%, the aim was to recruit 125 subjects to each arm of the trial (i.e. 375 patients in total).

2.6. Statistical analyses

Continuous and scale outcomes were analysed by means of linear mixed effect models for repeated measures adjusted for baseline measurements including all patients with at least one post-baseline measurement [41]. Frequencies were compared using χ^2-tests. For small samples, exact confidence intervals for proportions were computed. The resulting *p*-values were not corrected for multiple testing. All analyses were carried out using SAS version 9.

2.7. Qualitative studies

Qualitative studies were conducted in parallel with the main RCT, examining: (a) patient experience and benefits of the service, issues around the provider; and (b) telecare supporters' experience of training and intervention delivery, satisfaction and support, as well as any personal benefits gained [35]. Approximately one third of patients allocated to the intervention groups (35 allocated to PS group and 17 allocated to the DSN group) were invited to participate in semi-structured interviews, ensuring a widespread of age, gender, diabetes duration, baseline self-efficacy scores and baseline HbA1c. A total of 21 patients (14 from the PS group and 7 from the DSN group) were interviewed. Semi-structured interviews were also conducted with supporters 1 year into the intervention (seven peer supporters and three DSNs). Interviews were recorded and transcribed verbatim. Data were imported in NVIVO and analysed using a thematic framework approach [42].

3. Results

3.1. Practice and patient population

Of the 43 practices who agreed to participate, 3 withdrew prior to commencing patient recruitment. Twenty-nine (72.5%) of the remaining practices successfully recruited patients, ranging in size from single-handed practices to partnerships of ten GPs (mean = 5.2, S.D. = 2.4). Patient recruitment was very variable (range 1–23 patients recruited/practice; mean = 7.7, S.D. = 5.9), reflecting differences in numbers of eligible patients, conflicting organisational requirements and practice nurse confidence in undertaking recruitment [43].

There were a total of 8976 patients identified as having diabetes (Fig. 1). Of these, 2625 (29.3%) had raised HbA1c (>7.4%), of whom 988 (37.6%) met all eligibility criteria. Practice staff invited 447 (41.3%) of these patients to participate in the study, and 231 (51.7%) consented to participate, of whom 90 (39%) were randomised to the PS group, 44 (19%) to the DSN group, 97 (42%) to the control group.

The mean (S.D.) baseline HbA1c for those randomised and those who did not consent to participate were 8.6% (1.3) and 8.8% (1.4), respectively; difference 0.17 (95% CI (−0.03; 0.38), $p = 0.10$). There was under-representation of minority ethnic groups. The low ethnic mix may, in part, reflect the requirement for adequate English to communicate over the phone. Four (1.7%) patients who entered the study were found to have HbA1c levels less than 7.4, and a further 7 (3%) patients withdrew during the study; among reasons given, concurrent illness, poor hearing, no longer willing to take part, too busy.

The baseline characteristics were similar for participants in each of the three groups (Table 1).

3.2. Impact of the intervention

Each telecare supporter delivered the intervention to a median of 10 patients (range 4–14) during the trial, with each patient receiving on average 4.5 calls (range 1–6 calls, S.D. = 2.2). From analysis of the telecare supporters' records, the calls lasted on average 9.5 min (range 1–37 min, S.D. = 6.3 min), with first calls lasting on average 13.3 min. The mean total contact time per patients was 54.8 min (range 9–130 min, S.D. = 25.5). There was no statistically significant difference between the call lengths of DSN and PS groups.

Table 2 shows the primary and secondary outcomes at baseline and 6 months. Self-efficacy (DMSES) improved for patients in the PS group, diabetes-related problems (PAID) reduced for those in the DSN group, and in all groups HbA1c improved. The outcomes for the three arms of the trial were then compared adjusted for baseline measurements (Table 3). There were no significant differences between groups in the primary outcome as measured by DMSES. Differences in terms of HbA1c levels were generally small and not statistically significant. There were also no

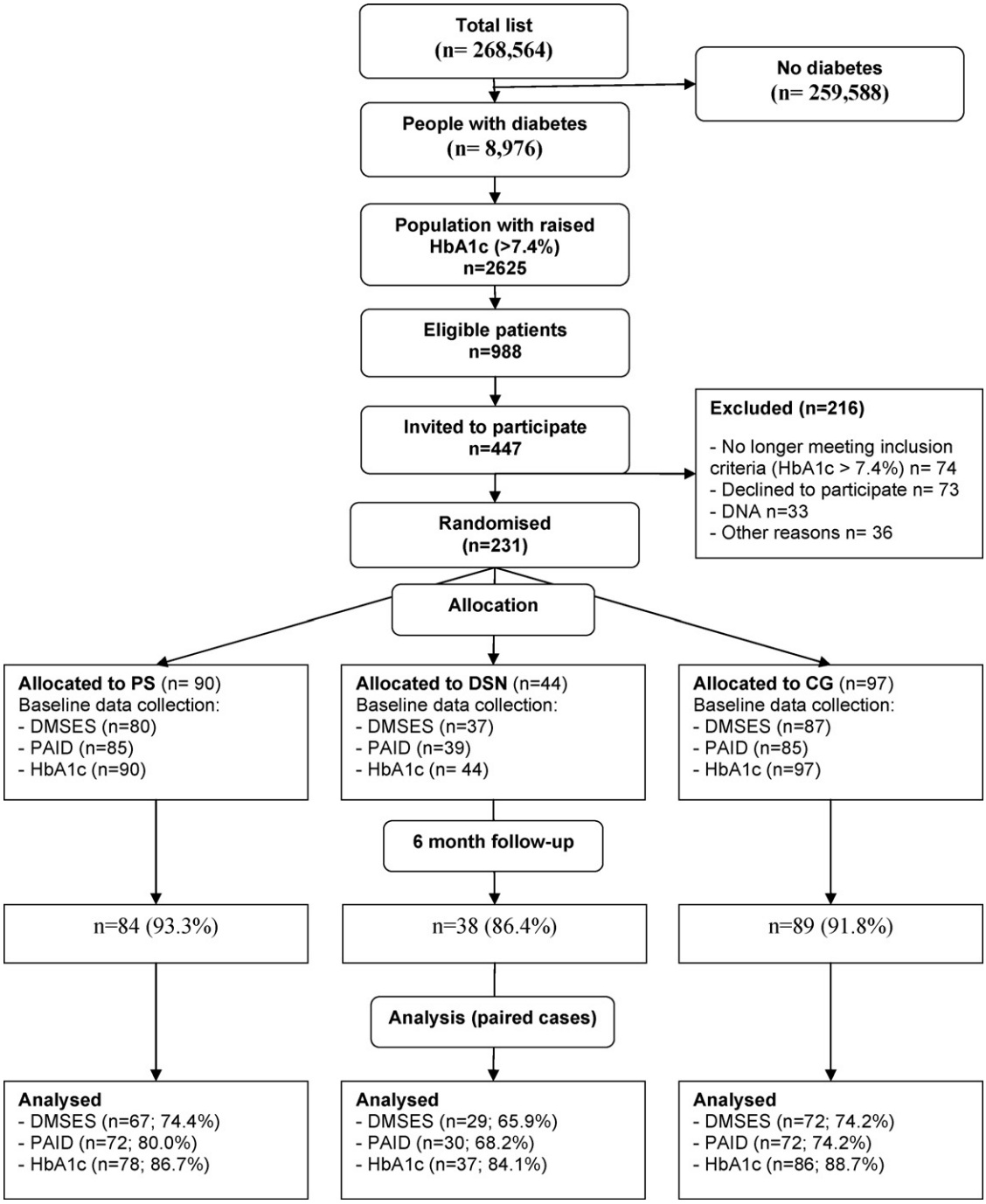

Fig. 1. Trial profile.

significant differences between groups in terms of diabetes distress as measured by PAID at baseline and follow-up.

We searched for subgroups (defined by their confidence in their ability to manage their diabetes) in which the intervention was more effective. However, no subgroups were identified with a statistically significant intervention effect for peer support compared to control or DSN support.

3.3. Patient and peer experience of the intervention

Satisfaction questionnaires were returned by 31 (70.5%) patients in the DSN group and 69 (76.7%) in the PS group. In all, 53/69 (77%) of PS group respondents compared to 29/31 (94%) of those in the DSN group said they would recommend the telecare support to other patients ($p = 0.04$). For the PS group, there was trend ($p = 0.06$) towards acceptability varying according to the telecare supporter to which patients were assigned (Fig. 2). There were no differences between groups in relation to other aspects of satisfaction, such as the telecare supporter's ability to listen.

The qualitative data confirmed that patients generally found the telephone support to be a useful addition to routine care. The calls were perceived as professional, friendly and non-judgmental, and an opportunity to share information with 'fellow sufferers'. They provided an opportunity to reflect on their diabetes management and establish goals, such as healthier diet, increased exercise levels, and better medication adherence. As a result, they

Table 1
Participants' demographic characteristics.

	Missing (PS/DSN/CG)	PS group, $n = 90$	DSN group, $n = 44$	CG, $n = 97$
Gender (male/%)	2/2/8	49 (55.7)	22 (52.4)	57 (64.0)
Age group (n/%)	2/2/8			
Under 50 years-old		17 (19.3)	6 (14.3)	12 (13.5)
51–69 years-old		46 (52.3)	25 (59.5)	54 (60.7)
Over 70 years-old		25 (28.4)	11 (26.2)	23 (25.8)
Ethnicity (n/%)	3/2/9			
White British		81 (93.1)	41 (97.6)	81 (92.1)
White Irish		3 (3.5)	0	2 (2.3)
White other		1 (1.2)	0	1 (1.1)
Indian		0	1 (2.4)	3 (3.4)
Black Caribbean		1 (1.2)	0	1 (1.1)
Black other		1 (1.2)	0	0
Diabetes duration (n/%)	2/2/9			
Less than 1 year		16 (18.2)	6 (14.3)	12 (13.6)
1–15 years		65 (73.9)	30 (71.4)	72 (81.8)
Over 15 years		7 (8.0)	6 (14.3)	4 (4.6)
Smoking status (n/%)	5/5/11			
Non-smoker		71 (83.5)	38 (97.4)	74 (86.1)
Smoker		14 (16.5)	1 (2.6)	12 (14.0)
Treatment changes advised by GP/nurse at the time of recruitment (n/%)	0/0/0			
Diet		10 (11.1)	6 (13.6)	11 (11.3)
Exercise		1 (1.1)	–	–
Diet and exercise		19 (21.1)	10 (22.7)	24 (24.7)
Medication		46 (51.1)	25 (56.8)	47 (48.5)
Medication and diet		10 (11.1)	–	8 (8.2)
Medication, diet and exercise		4 (4.4)	3 (6.8)	7 (7.3)

Table 2
Primary and secondary outcomes at baseline and 6 months.

Outcome	Group	Baseline			6 months			Difference	
		n	Mean (min, max)	S.D.	n	Mean (min, max)	S.D.	n	Mean (95% CI)
DMSES, Self-efficacy range: 0 (low)–150 (high)	PS	80	108.0 (7.5, 150.0)	30.4	73	116.3 (54.0, 150.0)	25.7	67	7.3 (2.6, 12.1)
	DSN	37	109.9 (61.1, 143.0)	21.5	33	109.4 (39.0, 148.0)	27.4	29	3.8 (−5.2, 12.9)
	CG	84	106.4 (13.9, 150.0)	30.8	77	111.3 (18.0, 150.0)	30.7	72	3.1 (−4.1, 10.3)
PAID, Problems/distress range: 0 (low)–80 (high)	PS	85	14.6 (0.0, 57.0)	12.7	75	13.3 (0.0, 71.0)	14.1	72	−1.7 (−4.1, 0.7)
	DSN	39	22.7 (1.0, 66.0)	18.8	33	17.5 (1.0, 64.2)	15.0	30	−5.1 (−9.2, −1.0)
	CG	85	19.8 (0.0, 56.0)	15.5	77	13.0 (0.0, 68.0)	14.7	72	−4.5 (−7.4, −1.6)
HbA1C	PS	90	8.4 (6.3, 13.2)	1.1	78	8.0 (5.2, 12.1)	1.5	78	−0.4 (−0.8, −0.1)
	DSN	44	8.9 (7.4, 16.1)	1.5	37	7.9 (5.9, 10.1)	0.9	37	−0.9 (−1.6, −0.3)
	CG	97	8.7 (5.6, 14.7)	1.3	86	7.9 (5.9, 12.1)	1.1	86	−0.8 (−1.0, −0.5)

felt more confident in managing their diabetes. Most patients felt they established a good rapport with their peer supporter and were able to share experiences and tips on how to manage their diabetes. They valued their supportive comments and the positive feed-back, while acknowledging the limitations of peer support with regard to the provision of information and advice. While most patients acknowledged the potential benefits of such a service, especially when a person is first diagnosed with diabetes, those who felt that their diabetes was already under adequate control did not feel that they derived particular benefits (Fig. 3).

Interviews with telecare supporters indicated that they all enjoyed helping others and derived personal benefits, such as improved knowledge, attitudes, skills, social relationships, and insights. They valued the positive feedback from patients and the improvements that were reported to have occurred as a result of the telephone support. DSNs reported having made changes in their professional practice as a result of their participation in the trial; for example, using listening skills to identify areas of concern and setting short-term goals with patients. Peer supporters reported that their confidence to self-manage their own diabetes had increased, and that this had helped them implement lifestyle changes, such as regular exercise, healthier diet and weight loss. Negative aspects of the telecare supporter role included difficulties associated with finding convenient times for calls, the number of

Table 3
Differences between the PS-CG, PS-DSN, and DSN-CG groups and at 6 months in the primary outcome (DMSES) and secondary outcomes (PAID and HbA1c).

Outcome	Comparison	Estimate	95% CI	p-Value*
DMSES	PS-CG	4.17	(−3.41; 11.74)	0.28
	PS-DNS	3.78	(−6.08; 13.64)	0.45
	DSN-CG	0.38	(−9.43; 10.19)	0.94
PAID	PS-CG	1.86	(−1.47; 5.19)	0.27
	PS-DNS	0.55	(−3.84; 4.93)	0.81
	DSN-CG	1.31	(−3.04; 5.66)	0.55
HbA1c	PS-CG	0.18	(−0.18; 0.54)	0.33
	PS-DNS	0.18	(−0.28; 0.63)	0.44
	DSN-CG	−0.00	(−0.45; 0.45)	1.00

CI: confidence interval.
* No statistically significant differences.

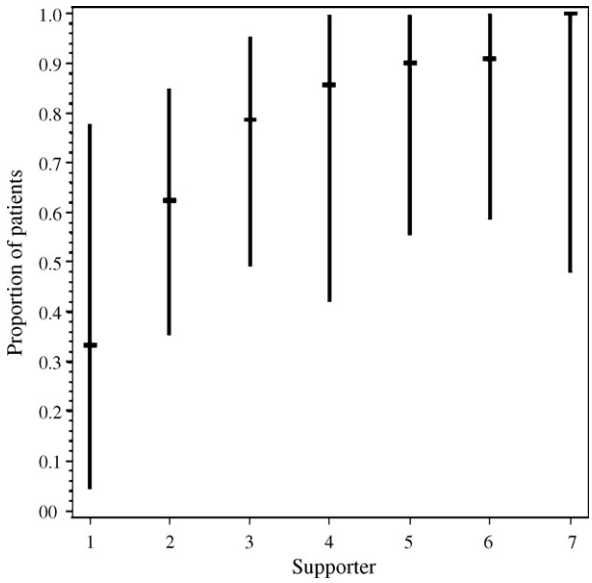

Fig. 2. Proportion of patients (with 95% CI) that would recommend the Telecare peer-support at month 6 by peer-supporter.

What was your opinion on the telephone support service?

- **Satisfaction with support:**

- *'I found it very helpful and understanding'. (P5)*
- *'It was nice to know that there was somebody there that was going through the same things as I was'. (P3)*
- *'It made me aware of my condition in that it made me aware of the needs of where my diabetes was getting out of control, and I think it focused my mind on those areas'. (P4)*

- **No additional support was felt necessary:**

- *'It's a good idea but I've sort of got it all under control myself'. (P6)*
- *'I told them I was quite happy with everything. I think [my diabetes] will probably get worse but it's not good worrying about it now'. (P8)*

Have you implemented any lifestyle changes as a result of the telephone support?

- **Changes in diabetes management:**

- *'I'd started to exercise and it made me realise how important diet was'. (P2)*
- *'I had started a walking regime'. (P5)*

Fig. 3. Study participants' views of the telephone support service.

What were the most rewarding calls?

- People who were setting goals and implementing changes:
  - *'I had [...] someone who kept saying that they would do some exercise and kept putting it off, then finally sort of admitted that they had been bluffing and they then started doing regular exercise.' (PS2)*
  - *'I gave him the guidance to go and find out for himself more information about diabetes [...] He went to his practice nurse and he started getting regular educational sessions.' (D3)*

What were the most challenging calls?

- People who did not want to make any changes: *'I've put my name down because I have to do it. I have no problems.' (P6)*

What did you enjoy most about participating in the study?

- **Satisfaction with helping other people:**

'It's hugely satisfying when you've got somebody on the end of the phone, who's... the sort of third time that you've telephoned them, and she tells you that she's losing weight and she's achieving what she's setting out to achieve' (PS7)

- **Learning that lead to changes in DSNs professional practice**:

'I was one that was always trying to tell the patient what to do, rather than getting the patient to see why they needed to make a change, [...] whereas now I word everything in a different way and try and listen to [...] their problems and then try and guide them through it.' (D1)

Fig. 4. Telecare supporters' views of the telephone support service.

attempts needed to make a call, and for some patients, challenges with the call content (e.g. keeping people focused, resistance to setting goals and making changes) (Fig. 4).

4. Discussion and conclusion

4.1. Discussion

In this exploratory trial, most patients found telephone support, whether provided by a peer or by a diabetes specialist nurse, to be a useful addition to routine care and said that they would recommend it to others. However, there was no evidence of such support being associated with improved outcome. Although HbA1c levels significantly improved in all three arms of the trial, this improvement occurred alongside the introduction of national financial incentives (QOF targets) to treat diabetes patients to target [44]. This meant that all patients were likely to have been receiving more intensive diabetes care during the course of the study than had previously been the case.

Most participants valued having someone they could talk to about diabetes who could offer them support and encouragement [45–47]. They varied in the extent to which they engaged with goal-setting. Those who were satisfied with their diabetes control were less likely to set goals and tended to use the telephone calls for casual support rather than a therapeutic interchange. This suggests that receptiveness to the need to improve diabetes control may be an essential prerequisite for this type of intervention to have a significant impact.

There was also some evidence that the opportunity to seek support and encouragement from a nurse was more highly valued than seeking such support from a peer. While the DSNs' credibility might be explained by their professional knowledge, manner and breadth of experience, this may also underline the need for careful selection, adequate training and ongoing support for the peer supporters.

The data revealed the importance for peer supporters of 'sharing their experience' with other people living with diabetes, reflecting the therapeutic role of 'making a connection' with others living with similar conditions [48]. Consistent with previous studies [12,14,25], adequate training, ongoing support and reinforcement are key elements needed to develop and maintain the supporters' sense of competency and to prevent feelings of isolation. Some studies point to the risk of burnout, particularly among peer supporters [23,48,49], and the risk of role strain could lead over time to a decrease in satisfaction. The qualitative interviews conducted with peer supporters revealed concerns related to challenging calls (e.g. patients unwilling to make any changes), frustration due to the number of attempts sometimes needed to be able to contact a person, and to personal health problems (e.g. asthma attack) impairing the ability to make calls.

4.2. Methodological considerations

A number of methodological issues emerged. Firstly, the difficulties encountered with patient recruitment and follow-up resulted in a failure to achieve the recruitment targets. Recruiting GP practices to participate in the study took much longer than anticipated, reflecting the intricacies of undertaking a pragmatic complex trial within the context of the conflicting priorities of everyday general practice. The majority of practices declining participation in the study mentioned workload demands or staff shortages as the main reason. Even for those that agreed to participate, patient recruitment proved difficult due to limited time during patient consultations, and a perception that patients lacked interest in the trial; for example, because of apprehension

about talking to strangers over the phone, and being too busy [43]. Although it was intended that patients should only be excluded from being invited to participate in the study where there were objective reasons for so doing, in practice many GPs and practice nurses were reluctant to invite patients to participate who they thought were unlikely to value additional support.

Secondly, assessing patients' readiness to implement diabetes-related behaviour change and self-confidence in their diabetes management prior to inclusion in the trial would have been helpful. As the purpose of the intervention was to encourage and motivate improved self-management that had been recommended in general practice it was assumed that the referring healthcare professional would have already persuaded patients of the need to make such change. However, it became clear that some patients were resistant to setting goals or were in state of denial or avoidance about the need to improve their diabetes self-management.

Thirdly, further consideration needs to be given to the training and preparation that is provided to telecare supporters. For example, it might have been beneficial to have included in the supporters' training programme more about the concepts of readiness to make changes and measures of confidence in doing so, alongside the development of active listening skills.

In addition, further consideration needs to be given to the choice of outcome measures for this type of intervention. The DMSES has been developed as a tool to measure self-efficacy in diabetes, but is still only partially validated [39]. The DMSES scale may have failed to identify the extent to which self-efficacy changed. Likewise, the levels of diabetes distress (as measured by PAID) were low in all three groups even at baseline. Hence there was little scope for demonstrating improvement in either outcome measurement. Additional outcome measures that could be used include the recently described Diabetes Obstacles to Adherence questionnaire [50].

Finally, the statistical analysis did not take into account any clustering effects for patients embedded in different general practices. However, as the numbers of patients recruited from each practice was on average less than 10 and they were randomised post-recruitment this is unlikely to have had much of an effect. Of greater importance might be clustering associated with individual peer or DSN supporter effects. There was some evidence of individual supporter effects identified, and this should be taken into account in the design of any future trial. The anticipated effect in the power calculation was rather ambitious and smaller effect sizes should be considered when designing future studies.

4.3. Conclusions

In conclusion, this exploratory trial has demonstrated the feasibility of providing telecare support for patients with inadequately controlled type 2 diabetes in general practice. While there was evidence of high levels of acceptability, peer telecare support appeared to be less highly valued than telecare support from a diabetes specialist nurse. The study failed to demonstrate any impact on physiological or psychological measures of well-being. Given the methodological difficulties encountered, the findings need to be interpreted cautiously. A number of lessons can be learnt, however, which should inform future research related to peer telephone support.

4.4. Practice implications

Telecare peer support offers a means of providing encouragement and support to patients who have poorly controlled type 2 diabetes. The findings from this study indicate that it may not benefit all patients, and there is a need to develop further (i) the targeting of the intervention (e.g. focusing on patients who are newly diagnosed with diabetes rather than those who have had the condition for several years; those who lack confidence with managing their diabetes; those who feel in need of more support), (ii) the training and on-going supervision of the telecare supporters to improve the consistency and fidelity of the intervention, and (iii) the extent to which advice in addition to support should be offered.

Conflict of interest

None declared.

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